COURSE STRUCTURE FOR B.TECH. Mechanical Engineering THIRD YEAR SEM – VI (w. e. f. 2024-25)

SEMESTER-VI (Subjects)				B.TECH. MECHANICAL ENGINEERING Sem-VI											
Sr. No	Category Code	Course Code	Course Name	Teaching Scheme					Exam Scheme						
									Theory			Practical		Total	
				L	T	P	C	Hrs/wk	MS	ES	CE	LE	LE/Viva	Marks	
1	OE		Open Elective 3 (From Other Department of FoET)	3	0	0	3	3	25	25	50	-	-	100	
2	PE	24ME335T	Mechanical Vibration	3	0	0	3	3	25	25	50	-	-	100	
		24ME336T	Additive Manufacturing						25	25	50	-	-	100	
		24ME337T	Production and Operation Management						25	25	50	-	-	100	
		24ME338T	Heat Exchanger Design						25	25	50	-	-	100	
3	PCC	24ME304T	Refrigeration and Air Conditioning	3	0	0	3	3	25	25	50	-	-	100	
4	PCC	24ME304P	Refrigeration and Air Conditioning Laboratory	0	0	2	1	2	-	-	-	25	25	50	
5	PCC	24ME305T	Machine Design -I	3	0	0	3	3	25	25	50	-	-	100	
6	PCC	24ME305P	Machine Design -I Laboratory	0	0	2	1	2	-	-	-	25	25	50	
7	PCC	24ME306T	Manufacturing Processes - II	3	0	0	3	3	25	25	50	-	-	100	
8	PCC	24ME306P	Manufacturing Processes - II Laboratory	0	0	2	1	2	-	-	-	25	25	50	
9	PCC	24ME307T	Robotics	3	0	0	3	3	25	25	50	-	-	100	
10	PCC	24ME307P	Robotics Laboratory	0	0	2	1	2	-	-	-	25	25	50	
			Total	**18**	**0**	**8**	**22**	**26**							

MS = Mid Semester ES = End Semester CE = Continuous Evaluation LW = Laboratory work LE = Laboratory Exam

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – VI

24ME335T					Mechanical Vibrations					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To learn theory of vibrations and characterize longitudinal, transverse and torsional vibration.
2. To formulate the equation of motion for single and multi-degree of freedom systems.
3. To formulate initial and boundary conditions for continuous system.
4. To design, analyse and interpret the significance of vibration isolators.

UNIT I: INTRODUCTION 11 Hrs.

Review of Single Degree of Undamped, Damped and Forced Vibration. Two Degree of Freedom System: Mode Shapes, Eigen Values and Eigen Vectors, Static and Dynamic Coupling, Semi-Definite System, Double Pendulum, Lagrange's Methods etc.

UNIT II: MULTI DEGREE OF FREEDOM SYSTEM 11 HRS.

Fourier Transform: Integral transform, Fourier integral theorem, Fourier sine and cosine integrals, Fourier transforms, Fourier sine and cosine transforms, Properties of Fourier transform, Convolution, Parseval's identity, Relationship between Fourier and Laplace transform

UNIT III: CONTINUOUS SYSTEMS 9 HRS.

Formulation of Initial Conditions and Boundary Conditions, Derivation of Longitudinal, Transverse and Torsional Vibrations of Shafts.

UNIT IV: VIBRATION CONTROL AND MEASUREMENT 9 HRS.

Vibration Control and Measurement: Vibration Isolators, Vibration Isolation with Rigid and Flexible Foundations, Active Vibration Control, Use of Vibration Absorbers, Vibration Pickups, Vibration Exciters, Signal Analysis, Dynamic Testing of Machines.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recall the basic concept of vibration and able to differentiate between damped and undamped vibrations.
CO2 : Analyze and understand single and two degree of freedom system.
CO3 : Formulate equation of motions of multi degree freedom system.
CO4 : Compare various methods and interpret the results.
CO5 : Construct equations of motion for continuous system.
CO6 : Identify and illustrate the practical applications of vibration absorbers, isolators and exciters.

TEXT/REFERENCE BOOKS

1. S.S. Rao, **"Mechanical Vibrations"**, Pearson Education.
2. W.T. Thomas, **"Theory of Vibration with Applications"**, Pearson.
3. L. Meirovitch, **"Fundamentals of Vibrations"**, Waveland Pr.
4. S.G. Kelly, **"Schaum's Outline of Mechanical Vibrations"**, McGraw-Hill.

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24ME336T					Additive Manufacturing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To provide fundamentals of additive manufacturing (AM) with recent development and applications
2. To study role of CAD models and CAM programming in AM with Reverse engineering.
3. To provide knowledge of different AM processes for various materials based on ASTM standards.
4. To develop experimental based learning with case study related to AM processes.

UNIT I: INTRODUCTION TO ADDITIVE MANUFACTURING **10 Hrs.**

Overview – History – Need- Classification of AM processes- fundamental engineering aspects. Applications of AM: AM applications in various Industries. Industry 4.0 and Digital and Smart manufacturing.

UNIT II: CAD FOR AM **10 Hrs.**

CAD model preparation – Part orientation and support generation – Model slicing –Tool path generation (preparatory (G) and miscellaneous (M) code generation) – Softwares for AM Technology. magics, mimics, STL file generation, part building errors.

UNIT III: AM PROCESSESS **14 Hrs.**

ASTM Classifications of AM processes, Fused deposition modelling (FDM)- Principle, process, advantages and applications Stereolithography (SLA)-Binder Jetting-Material jetting-Powder bed fusion AM processes involving sintering and melting- Principle, process, advantages and applications-Directed energy deposition-Sheet lamination: Principle, process, advantages and applications

UNIT IV: MATERIALS AND POST PROCESSING **8 Hrs.**

Different materials used in AM- polymers, metals, multiple materials, multifunctional and graded materials. Tooling and Manufacturing: Classification of RT Routes, RP of Patterns, Indirect RT: Indirect method for Soft and Bridge Tooling, Indirect method, Direct RT: Direct RT method for Soft and Bridge Tooling, Direct method for Production Tooling

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand fundamentals of Additive manufacturing (AM) with classification of AM process.
CO2 : Apply various designing and slicing techniques that enable AM and create programming for tool path.
CO3 : Understand fundamentals of polymer material based AM processes.
CO4 : Analyze the characteristics of Powder based AM process.
CO5 : Explain different materials used for building three dimensional AM components.
CO6 : Examine the various post processing techniques used in AM.

TEXT/REFERENCE BOOKS

1. Ian Gibson, David W. Rosen and Brent Stucker, "**Additive manufacturing technologies: rapid prototyping to direct digital manufacturing**", Springer.
2. C.K. Chua, K.F. Leong and C.S. Lim, "**Rapid prototyping: Principles and applications**", 3rd Edition, World Scientific.
3. Gebhardt A., **"Rapid prototyping"**, Hanser Gardener Publications.
4. Andreas Gebhardt, "**Understanding additive manufacturing: rapid prototyping, rapid tooling, Rapid manufacturing**", Hanser Publishers.

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24ME337T					Production and Operations Management					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To study significance on management of production operations.
2. To study the importance of forecasting and various techniques of forecasting
3. To develop skills of Inventory management, material requirement planning and project management
4. To evaluate various waiting line models, analysis of various queues and measuring queue performance

UNIT 1 FORECASTING AND INVENTORY 10 Hrs.

Operations Management: Introduction, Operations Management and Strategy, Tools for Implementation of Operations Forecasting: Introduction to Operations Management, operations strategy. Forecasting, time series methods of forecasting, causal methods of forecasting, measures of error, qualitative forecasting. Inventory management, inventory costs, and the basic EOQ model.

UNIT 2 MATERIAL REQUIREMENTS AND PLANNING 10 Hrs.

Material Requirements Planning (MRP), components of MRP, MRP logic, lot sizing in MRP systems (LFL, EOQ, POQ, LTC, LUC, etc.), introduction to ERP systems. Operations scheduling, loading, sequencing methods. Introduction to project management, Gantt charts, CPM, Activity on arrow /Activity on node networks, concept of slack, the critical path, probabilistic time estimates, project crashing. Introduction to manufacturing planning and control, brief about planning, routing, scheduling, loading, and dispatching

UNIT 3 SERVICE PROCESSES 10 Hrs.

Waiting line models, various types of queues, measures of queue performance, management of waiting lines. Introduction to facility planning, the facility location problem, factors influencing facility location, facility location models, process layouts, layout methods, computerized algorithms for facility layout, product layouts, fixed position layouts, cellular layouts, layout of service facilities.

UNIT 4 QUALITY MANAGEMENT 10 Hrs.

Introduction to quality management, cost of quality, quality management systems, concepts of TQM, quality tools, process capability, concept of six sigma, quality tools, control charts. Lean manufacturing, JIT, the Toyota Production System, waste elimination, push vs. pull systems, use of kanban.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Define the fundamentals of operations managements.
CO2 : Apply the principles of forecasting and inventory management
CO3 : Estimate cost and quantity of orders by materials requirements and planning techniques
CO4 : Analyse operations by using operations scheduling, Gantt charts and critical path methods.
CO5 : Evaluate various waiting line models, analysis of various queues and measuring queue performance
CO6 : Understand the concepts of quality management, six sigma and lean manufacturing.

TEXT/REFERENCE BOOKS

1. Chase, R. B., Ravi Shankar, Jacobs, F. R. and Aquilano, N. J., "**Operations & Supply Management"**, Tata McGraw Hill publishing company.
2. Buffa, E. S. and Sarin, R., "**Modern, Production and Operations Management**, John Wiley & Sons.
3. Martand Telsang, **Industrial engineering and production management**", PHI Learning
4. N. G. Nair, "**Production and operations management"**, Tata McGraw Hill publishing company.

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24ME338T					Heat Exchanger Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart the knowledge of construction, working and application of various heat exchangers
2. To perform the thermal and hydraulic design of various heat exchangers
3. To estimate the effect of geometric and operating parameters on the performance of heat exchangers

UNIT I: ADVANCED FUNDAMENTALS OF HEAT EXCHANGER DESIGN **10 Hrs.**

Classification of heat exchanger, selection of heat exchanger, review of LMTD & e-NTU method, multi-pass and cross flow heat exchanger, fouling, cleanliness factor, over surface design, techniques to control fouling, additives, rating and sizing problems, enhancement of heat transfer, performance evaluation of heat transfer enhancement technique. Real-world case studies.

UNIT II: SHELL AND TUBE HEAT EXCHANGER DESIGN **11 Hrs.**

Basic components of various STHE, TEMA standard, Classification as per TEMA standard, Thermal Design Theory for shell and tube heat exchanger, as per Kern and Bell-Delaware method, hydraulic design of shell and tube heat exchanger as per Kern method, Bell-Delaware method for Shell and tube heat exchanger design, introduction to HTRI software for STHE design

UNIT III: COMPACT HEAT EXCHANGER DESIGN **11 Hrs.**

Thermal design of plate-fin heat exchanger, fin and tube heat exchanger and plate heat exchanger, Estimation of pressure drop and hydraulic design of plate-fin, fin and tube, and plate heat exchange. Effect of geometric parameters and operating parameters on performance of heat exchanger. Design of double pipe heat exchangers. Selection of materials and innovative designs suitable for high-efficiency applications in aerospace, automotive, and renewable energy sectors, incorporating computational tools for design and analysis.

UNIT IV: REGENERATIVE HEAT EXCHANGER DESIGN **8 Hrs.**

Assumptions for Regenerator Heat Transfer Analysis, Λ-Π method for regenerator design, balance and symmetric regenerator design, unbalanced and unsymmetrical regenerator design, Influence of Matrix Material, Size, and Arrangement, Influence of longitudinal and transverse heat conduction. Applicability of phase change materials (PCMs) and additive manufacturing in regenerative heat exchangers, focusing on energy efficiency and innovation.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Identify and describe various heat exchangers, detailing their types, construction, components, operation, and applications
CO2 : Conduct thermal design for shell and tube heat exchangers, applying relevant methodologies and principles.
CO3 : Analyze the hydraulic design of shell and tube heat exchangers using diverse design approaches.
CO4 : Design plate-fin and fin-tube heat exchangers, assessing both their thermal and hydraulic performance.
CO5 : Implement design strategies for plate and double pipe heat exchangers, evaluating their operational efficiency.
CO6 : Execute regenerative heat exchanger design projects, assessing their effectiveness and performance.

TEXT/REFERENCE BOOKS

1. R.K.Shah, P. Sekulic, **"Fundamentals of Heat Exchanger Design"**, John Willey
2. Sadik Kakac, Hongtan Liu, **"Heat exchanger-selection, rating and thermal design**", CRC press
3. A.P. Frass, **"Heat exchanger design"**, Willey
4. Eric M. Smith, **"Advances in thermal design of heat exchangers"**, Willey.
5. VK Patel, VJ Savsani, MA Twahid, **"Thermal system design optimization"**, Springer Nature

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24ME304T					Refrigeration and Air-conditioning					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	-	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart the knowledge of construction, working and analysis of various refrigeration and air-conditioning system
2. To evaluate the performance of various refrigeration system and identify its application
3. To estimate the cooling/heating load and design the appropriate air-conditioning system

UNIT 1 VAPOUR COMPRESSION REFRIGERATION SYSTEM 14 Hrs.

Vapour Compression Refrigeration: Working principle and essential components of the plant, simple vapour compression refrigeration cycle - COP, Representation of cycle on T-S and P-h charts - Effects of various parameters on system performance - cycle analysis - Actual cycle, multistage compression system, and their analysis, multi-evaporator system and their analysis, cascade system. Industrial refrigeration system.Refrigerants: Desirable properties of refrigeration's; classification of refrigerants; ozone depletion global warming, secondary refrigerants, future industrial refrigerants, recent development.

UNIT 2 NON-CONVENTIONAL REFRIGERATION SYSTEMS 10 Hrs.

Air refrigeration: Introduction to Refrigeration; Necessity and applications; Bell-Coleman cycle, open and dense air systems, simple, boot strap, regenerative, and reduced air refrigeration system, comparison of various air refrigeration systems. Vapour Absorption Refrigeration: working principles; description and working of NH3- H20 system and LiBr- H20 system; three fluid absorption system and its salient features; Steam jet refrigeration system - working principle, basic components.

UNIT 3 PSYCHOMETRIC PROPERTIES AND PROCESS 11 Hrs

Psychometric – Composition of air and its effect on human comfort; psychometric terms – specific humidity, relative humidity percentage humidity and absolute humidity; temperatures – dry bulb, wet bulb and dew point; Psychometric processes: different types of psychometric processes sensible heating and cooling process, Latent heating and cooling process, heating with humidification, cooling with dehumidification, adiabatic cooling; air washer process; contact and bypass factor; apparatus dew pint; condition line; sensible heat factor; Requirement of human comfort.

UNIT 4 COOLING/HEATING LOAD CALCULATION, AIR CONDITIONING SYSTEMS & COMPONENTS DESIGN 10 Hrs

Cooling/Heating load calculation for various industrial & commercial applications; Selection of air-conditioning system, requirements of industrial air-conditioning system; Summer, winter and year-round air-conditioning system; Different method of duct design such as equal pressure drop, velocity reduction and static regain method

TOTAL HOURS: 45 HRS

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand the construction and working of various refrigeration & air-conditioning systems and list its applications.
CO2 : Analysed the various configuration of vapour compression refrigeration system and evaluate its performance.
CO3 : Investigate the performance of different conventional and nonconventional refrigeration systems
CO4 : Develop generalized psychometrics of moist air and apply to air-conditioning processes
CO5 : Design thermal comfort conditions with proper psychometric processes and evaluate its impact on human comfort, productivity, and health
CO6 : Estimate the cooling/heating load for different application and design the associate sub systems

TEXT/REFERENCE BOOKS

1. C. P. Arora, **Refrigeration and Air Conditioning"**, Tata McGraw Hill
2. R.J. Dossat, "**Principal of refrigeration**, John Willey
3. Manohar Prasad, "**Refrigeration and Air-conditioning"**, New age publishers
4. W.F. Stoker, "**Refrigeration and Air-conditioning"**, McGraw-Hill

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Semester – VI

24ME304P					Refrigeration and Air-conditioning Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	25	25	50

COURSE OBJECTIVES

1. Provide the knowledge of construction, working and operation of various refrigeration and air-conditioning equipments
2. To measure the performance parameter and evaluate the system performance of various refrigeration and air-conditioning equipments
3. To Identify the effect of operating parameters on the system performance

LIST OF EXPERIMENTS

1. Performance test on vapor compression refrigeration system to measure the COP of the system
2. Performance test on VCR system with continuous flow condition and measure the COP of the system
3. Performance comparison of the water-cooled and air-cooled condenser of VCR system
4. Performance test on the heat pump system
5. Performance test on CASCADE refrigeration system
6. Performance test on NH3-H2O vapor absorption refrigeration system
7. Performance test on three fluid vapor absorption refrigeration system
8. Performance test to obtain various psychrometric processes
9. Performance test on the air-conditioning system and measure the cooling capacity and COP of the system
10. Perform the cooling load calculation for the given space

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand the function of different components of refrigeration and air-conditioning systems with its construction and working.
CO2 : Evaluate the performance of compression-based refrigeration systems
CO3 : Analysed the effect of operating parameters on the performance of refrigeration and air-conditioning systems.
CO4 : Evaluate the performance of non-conventional refrigeration system
CO5 : Understand the different psychrometric terms and analyse the various psychrometric process
CO6 : Evaluate the performance of the air-conditioning system

TEXT/REFERENCE BOOKS

1. C. P. Arora, **Refrigeration and Air Conditioning"**, Tata McGraw Hill
2. W.F. Stoker, "**Refrigeration and Air-conditioning"**, McGraw-Hill

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Semester – VI

24ME305T					Machine Design - I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To gain familiarity with the concepts of design procedure under different types of loads.
2. To be able to design a pressure vessel as per industrial requirements.
3. To obtain the knowledge of designing power transmission systems.
4. To attain capability of designing various components of an Engine.

UNIT I: FAILURE RESULTING FROM STATIC AND VARIABLE LOAD 10 Hrs.

Design based on deflection and stiffness: Tension, compression and torsion based design, deflection in members, statistically indeterminate design problems, design of compression members, design for eccentric loading, design for shock and impact loading. **Failure resulting from static load:** Stress concentration, theories of failure for ductile and brittle material, contact stress **Failure resulting from variable load:** strain –life relationship, stress-life relationship, endurance limit, fatigue strength, notch sensitivity, torsional fatigue, fluctuating stress characterization, Goodman's line, Soderberg's line, Modified Goodman's line, surface fatigue strength, Design factor in fatigue.

UNIT II: DESIGN OF PRESSURE VESSELS 10 Hrs.

Types and applications of pressure vessels, Design of thin and thick cylindrical and spherical shells, Compounding of cylinders, Design of interference joints – press / shrink fitted assemblies, Design of cylinder covers, Cover plates.

UNIT III: DESIGN OF POWER TRANSMISSION SYSTEMS 10 Hrs.

Introduction and applications of power transmission systems. Types of power transmission devices, advantages and disadvantages, components of belt drive, mechanism of belt drive, Belt Materials, Types of belts and pulleys and their selection, Stresses in belts, Transmission efficiencies, Design/selection of flat belt drive, design/selection of V-belt drive, Design of Chain drives and wire ropes.

UNIT IV: DESIGN OF ENGINE PARTS 10 Hrs.

Components of I.C. engine, materials for engine components, working conditions, Forces and stresses acting on various engine components, Design of cylinder and cylinder liner, Design of piston, Design of Connecting Rod, Design of crank shafts, Design of crank pins.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recall fundamentals of failure of mechanical components related to static loading.
CO2 : Evaluate the life of component under variable loading.
CO3 : Solve the problem of cylinder for different thickness.
CO4 : Analyze the various stresses in pressure vessels.
CO5 : Compare different transmission devices such as belt and chain drives.
CO6 : Design the engine components such as cylinders, piston, connecting rod and crank shaft.

TEXT/REFERENCE BOOKS

1. V. Bhandari, **"Machine Design"**, Tata-McGraw Hill Publishers.
2. R. S. khurmi and J. K. Gupta, **"Machine Design"**, S Chand Publication
3. C. Sharma and K. Purohit, **"Design of Machine Elements"**, PHI Publishers.
4. J. Shigley, C. Mischke, R. Budynas, **"Mechanical Engineering Design"**, Tata-McGraw Hill Publishers.
5. R. Juvinall, K. Marshek, **"Fundamental of Machine Component Design"**, John Wiley and Sons Publishers.
6. R. Norton, **"Machine Design: An Integrated Approach"**, Pearson Education Publishers.

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24ME305P					Machine Design -I Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To define mechanical problem and learn mathematical formulation.
2. To learn Computational coding and CAD modelling.
3. To perform analysis and interpret the results.

LIST OF EXPERIMENTS

1. To study the basic of Geometric Dimensioning and Tolerancing (GD&T) in Machine Design.
2. To study design based on fatigue loads (variable loads).
3. To design a cylinder shells subjected to internal pressure (pressure vessel).
4. To design a spherical shells subjected to internal pressure (pressure vessel).
5. To design a Flat belts and pulleys drive system.
6. To design a V belts and pulleys drive system.
7. To design an IC engine cylinder subjected to internal stresses.
8. To design a connecting rod.
9. To design a crank shaft and crank pin.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Learn to utilize component catalogues and standard machine components effectively.
CO2 : Classify different types of analysis that can be performed on mechanical components.
CO3 : Prepare problem and Solve mathematical equations for simple interdisciplinary problem.
CO4 : Analyse the dimension of components using the analytical tools.
CO5 : Compare the results obtain using analytical and numerical modelling.
CO6 : Design and Develop Computational coding, CAD model and perform analysis for a mechanical components.

TEXT/REFERENCE BOOKS

1. Gene R. Cogorno, **"Geometric Dimensioning and Tolerancing for Mechanical Design"**, Tata-McGraw Hill Publishers.
2. V. Bhandari, **"Machine Design"**, Tata-McGraw Hill Publishers.
3. Rudra Pratap, **"Getting Started with MATLAB: A Quick Introduction for Scientists & Engineers"**, Oxford University Press.
4. Steven C. Chapra, **"Applied Numerical Methods with MATLAB for Engineers and Scientists"**, Tata McGraw-Hill Education.
5. Mary K. Thompson & John M. Thompson, **"ANSYS Mechanical APDL for Finite Element Analysis"** Butterworth-Heinemann.

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24ME306T					Manufacturing Process- II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To provide fundamentals of machining machine tools.
2. To study single point cutting tool with importance of different tool angles.
3. To provide knowledge of machining techniques with numerical and experimental understanding.
4. To understand gear indexing mechanisms and methods of gear manufacturing.

UNIT I: INTRODUCTION TO MACHINING AND TOOL GEOMETRIES 12 Hrs.

Need of machining – classification of machining. Introduction to machine tools and power drives. **Single point cutting tool:** Tool geometry– concept of rake and clearance angles –different systems of tool geometry –mechanism of chip formation- Merchant's circle analysis- essential properties and types of cutting fluids, machinability – tool life and tool failure.

UNIT II: LATHE, SHAPING, SLOTTING, AND PLANNING MACHINES 12Hrs.

Lathe: Lathe: Principle of working. Classification and specification of lathe machines – lathe accessories and attachments – various lathe operations - numerical and experimental study-effect of process parameters. **Shaping, slotting, and planning machines:** Kinematic systems – principle of working – classifications, specification and operations performed - machining time calculation.

UNIT III: MILLING MACHINES, GEAR INDEXING AND MANUFACTURING METHODS 08 Hrs.

Kinematic system – principle of working – classifications – specifications – operations performed – accessories – milling cutter – classifications of cutters. Introduction to gear indexing – methods of indexing – calculations of gear indexing methods– manufacturing methods of gears.

UNIT IV: DRILLING, BORING AND GRINDING MACHINES 08 Hrs.

Drilling and Boring machines: Kinematic system – principle of working – classifications – specifications – operations performed – machining time calculations. Classifications of grinding machines – specifications grinding wheels – mechanism of grinding – different bonds and abrasives–truing and dressing.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Study the fundamentals of machining and machine tools used in manufacturing.
CO2 : Analyze single point cutting tool geometry with role of different tool angels.
CO3 : Apply different principles of lathe machine with machining characteristic in manufacturing.
CO4 : Analyze the shaping, slotting and planning operations with machining time calculations.
CO5 : Understand fundaments of milling machines and evaluation of gear indexing methods.
CO6 : Explain drilling, boring, and grinding processes with different classifications.

TEXT/REFERENCE BOOKS

1. A. B. Chattopadhyay, **"Machining And Machine Tools"**, John Wiley & Sons Publisher.
2. Geoffrey Boothroyd, **"Fundamentals Of Metal Machining And Machine Tools"**, Crc Press.
3. R.K. Jain And S.C. Gupta, "Production Technology", Khanna Publishers.

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24ME306P					Manufacturing Process- II Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To understand the working principle and operations of conventional machining techniques such as lathe, milling, drilling, grinding, shaper and grinding machines.
2. To prepare a job by using lathe, milling, drilling, grinding, shaper and grinding machines. Different electronic circuits and their practical applications.
3. To understand the working principle and operations of non-conventional machining techniques.
4. To prepare a job by using of non-conventional machining techniques such as edm, wedm and ecm processes.

LIST OF EXPERIMENTS

1 To study the safety aspects of conventional and non-conventional machining operations.
2 To study the lathe machine components and different machining operations performed by lathe.
3 To prepare a job of required geometry by using lathe machine operations
4 To prepare a job of required geometry by using milling machine operations
5 To prepare a job of required geometry by using drilling machine operations
6 To prepare a job of required geometry by using shaper machine operations
7 To prepare a job of required feature by using grinding machine operations
8 To study the impact of various factors of edm process on performance measures through experimental trials.
9 To study the impact of various factors of wire-edm process on performance measures through experimental trials.
10 To perform electrochemical machining (ecm) machining operations.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Define fundamentals principles of metal cutting to practical applications using lathe machine operations
CO2 : Apply principles of metal cutting through milling and shaping machining processes.
CO3 : Analyze the effect of process parameters through drilling machine operations.
CO4 : Determine the effect of process parameters for grinding operations.
CO5 : Examine the effect of process parameters using ecm machining process.
CO6 : Evaluate the performance of edm, and wedm process

TEXT/REFERENCE BOOKS

1. Sharma PC, **"A Textbook Of Production Engineering",** S. Chand Publishing.
2. H.M.T., "**Production Technology**", Tata Mcgraw-Hill Education.
3. R.K. Jain And S.C. Gupta**, "Production Technology"**

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – VI

24ME307T					Robotics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand demonstrate the ability to design, analyse, and implement robotic systems.
2. To evaluate the kinematics and dynamics of robot manipulators.
3. To learn the basic of mapping, velocity, acceleration of robot manipulators.
4. To study the industrial robots from the view of Industry 4.0.

UNIT I: INTRODUCTION TO ROBOTICS 10 Hrs.

Introduction to Robotics, Laws of robotics, Robot anatomy, Links joints, DOF, Arm and wrist configuration, End effector, Cylindrical, Polar & Articulated configurations, Classification of Robotic manipulators, Co-ordinate frames, Mapping and Transformation – Rotation of vectors, Translation of vectors, Composite transformation also using software, Transformation matrix, Rotation matrix, Euler Angle representation, Fixed Angle representation.

UNIT II: KINEMATICS OF ROBOT 10 Hrs.

Description of Links and Joints, Frame assignments to the manipulators, Denavit-Hartenberg Notation, D-H Parameters using software, Kinematic relation between adjacent links, Manipulator Transformation Matrix, Forward Kinematics. Inverse Kinematics, Solvability of inverse kinematic model.

UNIT III: DIFFERENTIAL MOTION, TRAJECTORY PLANNING & CONTROL 10 Hrs.

Velocity propagation along links – linear and angular velocity analysis, manipulator Jacobian, Jacobian singularity, Jacobian in force analysis. Manipulator trajectory and analysis, Joint vs Cartesian space approach, motion planning, Trajectory planning. Open and closed loop control, Proportional, Integral & Derivative (PID) Control, Force and torque controls of manipulators.

UNIT IV: DYNAMICS OF ROBOT 10 Hrs.

Introduction to Dynamic modelling of manipulator, Lagrangian mechanics of manipulator dynamics, 2-DOF manipulator dynamic modelling, Lagrange-Euler formulation, Inverse dynamics, Software based dynamic modelling.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Comprehend the basics of robot manipulators.
CO2 : Understand the fundamentals of matrix transformation and rotation matrices.
CO3 : Assess the manipulator kinematic relationships using Denavit-Hartenberg parameters.
CO4 : Estimation of the differential motion - velocity analysis of robot manipulators.
CO5 : Evaluate the path and trajectory of robot manipulator.
CO6 : Formulate a dynamical model for robot manipulator.

TEXT/REFERENCE BOOKS

1. R K Mittal and I J Nagrath, **"Robotics and Control"**, TMH Edu. Pvt. Ltd..
2. Robert J. Schiling, **"Fundamentals of Robotics Analysis and control"**, PHI publishers.
3. John J. Craig, **"Introduction to Robotics: Mechanics and Control**", 3rd Edition, Pearson.
4. Siciliano, Bruno, Lorenzo Sciavicco, **"Robotics-Modelling, Planning and Control"**, Springer.

Pandit Deendayal Energy University, School of Technology
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Semester – VI

24ME307P					Robotics Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To comprehend the elementary terminology of robotic systems.
2. To accomplish the kinematic and dynamic modelling of several robotic manipulators.
3. To study the fundamental of programming language for different robots.

LIST OF EXPERIMENTS

1. Comprehend configurations and degree of freedoms of Robots using software.
2. Study the Forward Kinematical model of a manipulator using software.
3. Study the Inverse kinematical model of a manipulator using software.
4. Formulate the Dynamical model of Robotic system using software.
5. Interfacing of Input-Output devices using Open-source developmental board - Arduino.
6. Interfacing of Input-Output devices using Open-source developmental board - Raspberry Pi.
7. Programming of Computer Vision using software.
8. Proportional-Integral-derivative (PID) control of an inverted pendulum system.
9. Learn the procedure of manipulators using teach pendent.
10. Build a Robotic system - group wise student mini projects and Case studies.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recognize the elementary terms of robotic manipulators.
CO2 : Understand the forward and inverse kinematical model of a robot.
CO3 : Develop robot programs for interfacing Arduino and Raspberry Pi.
CO4 : Formulate a computer program of robot vision.
CO5 : Analyse and understand the working of industrial robot manipulators.
CO6 : Design state-of-the-art robots as case studies.

TEXT/REFERENCE BOOKS

1. R K Mittal and I J Nagrath, **"Robotics and Control"**, TMH Edu. Pvt. Ltd..
2. Robert J. Schiling, **"Fundamentals of Robotics Analysis and control"**, PHI publishers.
3. John J. Craig, **"Introduction to Robotics: Mechanics and Control"**, 3rd Edition, Pearson.
4. Siciliano, Bruno, Lorenzo Sciavicco, **"Robotics-Modelling, Planning and Control"**, Springer.